(Please write your Enrollment Number) Enrollment No. 04801172022

End-Term Examination

(CBCS)(SUBJECTIVE TYPE)(OffLine)

Course Name: B.Tech./CSE-AI, Semester: 5

(December, 2024)

Subject Code: BAI 305	Subject: Deep Learning-I
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(5*4=20)	CO Mapping
	a) "Perceptron Learning Algorithm is applicable on multi-classification problem." Justify your answer with appropriate reason.		CO1
	b) Differentiate between L1 and L2 regularization.		CO2
	c) Why is sigmoid function not appropriate for hidden layers of CNN? Explain the problems in details.		CO3
	d) Explain the different strategies for weight initialization of a DL model.		CO4
	UNIT I		CO Mapping
Q2	Consider a three-input perceptron with weight vector a [1,3,4] with bias as -1. For the input vectors of [0.1, -4, -5] with output label as 1 and [2, 2.5, 0.6] with output label as 0, what will be the updated weights values after two iterations of perceptron learning algorithm?	(10)	CO1
Q3	MP-neuron units and perceptron units are suitable for linear separable Boolean functions. Identify which one is better and justify your answer with appropriate reason	(10)	CO1
	UNIT II		CO Mapping
Q4	Consider a two-input perceptron with sigmoid activation function and weight vector as [0.1, 4] and bias of 0.5. For the input vector of [7.2, -0.5] and output label as 1 and cost function as $(y^2+y^*)^2$, where 'y' is the target output and 'y*' is the estimated output. What will be the updated weights values after two iterations of gradient descent algorithm (learning rate as 0.2)?	(10)	CO2
Q5	Differentiate between Overfitting vs Underfitting. Explain the dropout and data augmentation in detail in context of ANN.	(10)	CO2
	UNIT III		CO Mapping
Q6	"ReLU is suitable for CNN while TanH is suitable for RNN". Justify this statement with appropriate reason. Also discuss the other activation functions available for deep learning models.	(10)	CO3
Q7	Explain the LSTM and GRU unit along with architectural difference.	(10)	CO3
	UNIT IV		CO Mapping
Q8	"Convolutional Neural Network is better than Recurrent Neural Network for Image Classification". Identify the correctness of the above statement and justify with appropriate reason.	(10)	CO4
Q9	Explain the VGG16, ResNet, and InceptionNet along with their architecture in detail.	(10)	CO4